

Optional Work on Portfolio Optimization

Consider a very simple situation in which we have a financial model with only two times: time 0 and time 1. At time 0, an agent invests an initial amount $X_0 > 0$. There is a single risky asset, namely a stock S , and a bank account with risk-free rate r . The stock can be purchased or sold short (in any amounts, including fractional shares) with no transaction costs and we can borrow or invest any amount of money we want at the bank. The same rate r applies to loans and deposits. We use single-period compounding. In particular, if an amount $A > 0$ is invested (or borrowed) at time 0 the investor will have $A(1 + r)$ in the account (or owe the bank $A(1 + r)$) at time 1. The agent wants to decide how to allocate the initial capital between stock and the bank account in order to achieve an *optimal result*. The capital X_1 of the portfolio at time 1 is unknown at time 0 (it is a random variable) so we cannot maximize X_1 . It might seem natural to maximize $\mathbb{E}[X_1]$ (the expected value of the terminal capital), but this is not a good idea because of *risk aversion*. For most people the pain caused by the loss of a given amount δ is greater in magnitude than the joy experienced by a gain of the same amount δ ; gains and losses should not be treated symmetrically. This is where *utility functions* come into play.

Let's assume here that only strictly positive terminal capitals will be considered. A utility function is a smooth function $U : (0, \infty) \rightarrow \mathbb{R}$ satisfying

$$U'(x) > 0 \text{ for all } x > 0 \text{ (more money is always better),}$$

$$U''(x) < 0 \text{ for all } x > 0 \text{ (this expresses risk aversion).}$$

The investor will maximize $\mathbb{E}[U(X_1)]$, rather than $\mathbb{E}[X_1]$. The condition $U'' < 0$ means that the utility function is concave. By Jensen's inequality the expected utility $\mathbb{E}[U(X_1)]$ is less than $U(\mathbb{E}[X_1])$. The utility of the expected terminal capital is greater than the expected utility of the terminal capital. That is, the investor will be happier to receive a nonrandom payout of amount $\mathbb{E}[X_1]$ than any random payout having the same expected value because the nonrandom payment has no risk.

To simplify things further here, let's assume that there are only two possible values of the stock price S_1 at time 1. A fair coin (probability of heads = $\frac{1}{2}$) will be tossed at time 1. If the result is H , we will have $S_1 = uS_0$. (Here S_0 is the initial stock price, S_1 is the stock price at time 1, and u is a given number called the up factor.) If the result of the coin toss is T we will have $S_1 = dS_0$, where d is a given number called the down factor. In order for this model to be reasonable we must have

$$0 < d < 1 + r < u.$$

This is the one-period binomial model. Although it is very simplistic, it can be used to build much more interesting models. (In general the probability of a head does not need to be $\frac{1}{2}$, but it is convenient to assume so here.)

In order to connect this simple problem with actual markets here, we take

$$(*) \quad u = 1 + \alpha + \sigma, \quad d = 1 + \alpha - \sigma,$$

where α is the expected return of the stock and $\sigma > 0$ is the volatility.

To start off, let's take $r = .06$, $\alpha = .12$, and $\sigma = .18$ which gives us $u = 1.3$ and $d = .94$. You should experiment with other values on your own. Let Γ be the total initial capital and y be the dollar amount of stock in the portfolio at time 0. (This means that $\Gamma - y$ will be the amount invested in the bank at time 0. (This amount could be negative – the investor might borrow money to buy stock. If the investor short sells the stock then we will have $\Gamma - y > \Gamma$.) The capitals at time 1 for H and T will be given by

$$X_1^H(y) = (\Gamma - y)(1 + r) + uy, \quad X_1^T(y) = (\Gamma - y)(1 + r) + dy.$$

The expected utility of the terminal capital (as a function of y) is therefore given by

$$(*) \quad g(y) = \frac{1}{2}U((\Gamma - y)(1 + r) + uy) + \frac{1}{2}U((\Gamma - y)(1 + r) + dy).$$

Let's use utility functions of the following form: We take a number $\beta \geq 0$. The derivative of our utility function is given

$$(**) \quad U'(x) = \frac{1}{x^{\beta+1}}, \quad x > 0.$$

(Notice that $\delta = 0$ corresponds to a logarithmic utility function and for $\beta > 0$ the utility function is $U(x) = -\frac{1}{\beta}x^{-\beta}$.) This class of utility functions is widely used. We can discuss other utility functions later if there is sufficient interest.

(a) The return ρ of the stock is given by

$$\rho = \frac{S_1 - S_0}{S_0}.$$

Verify that if u and d are given by (*) and the coin is fair (probability of heads equal to $\frac{1}{2}$) then we have

$$\mathbb{E}[\rho] = \alpha, \quad \text{Sd}(\rho) = \sigma.$$

(Here $\text{Sd}[\rho]$ is the standard deviation of ρ .)

(b) Find the value of y that maximizes $g(y)$ when $r = .06$, $u = 1.3$, $d = .94$ and $\beta = 0$ in (**). Your answer should involve specific numbers and Γ .

- (c) For a general value of $\beta > 0$ in (**) find the value of y that maximizes $g(y)$ when $r = .06$, $u = 1.3$, $d = .94$. Your answer should involve specific numbers, Γ and β . What observations can you make?
- (d) For general r , α , σ , and $\beta \geq 0$ in (**), let $u = 1 + \alpha + \sigma$, $d = 1 + \alpha - \sigma$ and find the value of y that maximizes $g(y)$. What observations can you make? In particular, under what conditions will the optimal portfolio have a short position in the stock?